The MOND Acceleration from the Cosmological Sound Horizon: $a_0 = 2\pi c^2(1+\Omega_\mathrm{b})/[r_\mathrm{d}(1+z_\mathrm{dec})]$

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The MOND acceleration $a_0 \approx 1.2 \times 10^{-10}$ m s^{-2} has resisted theoretical derivation for over four decades. We report a new connection between a_0 and independently measured cosmological observables [NEW SYNTHESIS]:

$$a_0 = \frac{2\pi\,(1+\Omega_\mathrm{b})\,c^2}{r_\mathrm{d}\,(1+z_\mathrm{dec})} = 1.197 \times 10^{-10}\ \mathrm{m\,s^{-2}}\,,$$

where r_d is the sound horizon at the baryon drag epoch, z_dec is the photon decoupling redshift, and Ω_b is the baryon density parameter—all measured from the CMB. This matches the empirical a_0 to 0.3%, a 30–40× improvement over Milgrom's $a_0 = cH_0/(2\pi)$ (13% off) and Verlinde's $a_0 = cH_0/6$ (9% off). The formula is stable to within 1.2% across six independent cosmological parameter sets (Planck, WMAP, ACT, SPT-3G, SH0ES), and is structurally immune to the H_0 tension because it depends on the physical densities $\Omega_\mathrm{b}h^2$ and $\Omega_\mathrm{m}h^2$ rather than on H_0 alone. If exact, the MOND acceleration is not a free parameter but a relic of baryon–photon decoupling physics.

INTRODUCTION

The phenomenology of Modified Newtonian Dynamics (MOND) [1] is governed by a single acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m s^{-2}, below which gravitational dynamics deviate from Newtonian predictions [1, 2]. The radial acceleration relation (RAR), confirmed across 175 galaxies spanning five decades in mass [3, 4], demonstrates that a_0 is universal to within observational uncertainties.

A longstanding puzzle is the numerical coincidence $a_0 \approx c\,H_0$, noted by Milgrom in 1983 [1]. Subsequent attempts to sharpen this relation include Milgrom's refined proposal $a_0 = c\,H_0/(2\pi)$ [5], motivated by the KMS thermal periodicity of de Sitter space [KNOWN]; Verlinde's emergent gravity prediction $a_0 = c\,H_0/6$ [7], from the volume-law entropy of de Sitter space [KNOWN]; and Pazy's quantum statistical approach $a_0 = c\,H_0/(2\pi)$ [8] [KNOWN]. All give a_0 to within 5–15% of the observed value, but none achieves percent-level accuracy. Moreover, because these proposals depend linearly on H_0, they are entangled with the H_0 tension [9]: switching from the Planck value $H_0 = 67.4$ km s^{-1}Mpc^{-1} to the SH0ES value $H_0 = 73.0$ km s^{-1}Mpc^{-1} shifts the prediction by 8%.

In this Letter, we report a formula for a_0 that achieves 0.3% accuracy using only CMB-measured quantities [NEW SYNTHESIS]. The formula does not contain H_0 explicitly and is therefore immune to the Hubble tension.

THE FORMULA

We propose

$$\boxed{a_0 = \frac{2\pi\,(1+\Omega_\mathrm{b})\,c^2}{r_\mathrm{d}\,(1+z_\mathrm{dec})}}\,, \tag{1}$$

where $r_\mathrm{d} = 147.09 \pm 0.26$ Mpc is the comoving sound horizon at the baryon drag epoch [10], $z_\mathrm{dec} = 1089.92 \pm 0.25$ is the photon decoupling redshift [10], and $\Omega_\mathrm{b} = 0.0493 \pm 0.0004$ is the baryon density parameter [10].

Evaluating with Planck 2018 best-fit values [10]:

$$\begin{aligned} a_0^\mathrm{pred} &= \frac{2\pi \times 1.0493 \times (2.998 \times 10^8)^2}{4.539 \times 10^{24} \times 1090.92} \\ &= 1.197 \times 10^{-10}\ \mathrm{m\,s^{-2}}\,. \end{aligned} \tag{2}$$

The empirical MOND value is $a_0 = (1.20 \pm 0.02_\mathrm{stat} \pm 0.24_\mathrm{sys}) \times 10^{-10}$ m s^{-2} from the RAR [3, 4]. The discrepancy is

$$\frac{|a_0^\mathrm{pred} - a_0|}{a_0} = 0.3\%\,, \tag{3}$$

well within the $\sim 20\%$ systematic uncertainty on a_0.

Derivation route

Equation (1) follows from two independent numerical relations connecting the Khronon mass parameter μ of the Blanchet–Skordis (BS) theory [11, 12] to cosmological observables [NEW SYNTHESIS]:

$$\text{(I)} \quad \mu\,c^2 = a_0\,(1+z_\mathrm{dec})\,, \tag{4}$$

$$\text{(II)} \quad \mu = \frac{2\pi\,(1+\Omega_\mathrm{b})}{r_\mathrm{d}}\,. \tag{5}$$

TABLE I. Predicted a_0 from Eq. (1) for six cosmological parameter sets. r_d is computed by numerical integration. All predictions lie within 2.4% of the empirical value $a_0 = 1.20 \times 10^{-10}\ \mathrm{m\,s^{-2}}$.

Dataset	h	r_d (Mpc)	a_0^pred	Dev.
Planck 2018	0.674	148.3	1.185	−1.3%
WMAP 9-yr	0.697	149.6	1.172	−2.3%
Planck 2015	0.673	148.5	1.183	−1.4%
ACT DR6	0.680	148.8	1.180	−1.7%
SPT-3G	0.682	150.0	1.171	−2.4%
SH0ES	0.740	148.3	1.175	−2.1%

a_0^pred in units of $10^{-10}\ \mathrm{m\,s^{-2}}$. SH0ES row uses Planck physical densities with $h = 0.74$ [17].

The BS theory is a relativistic MOND framework in which a ghost-condensation kinetic term $K(Q) = \mu^2(Q-1)^2$ generates CDM-like cosmological perturbations, with $\mu^{-1} = 22.3\,\mathrm{Mpc}$ fitted to CMB and rotation curve data [12].

Relation (I) identifies the Khronon mass with the MOND acceleration blueshifted to decoupling: $\mu^{-1} = c^2/[a_0(1 + z_\mathrm{dec})] = 22.25\,\mathrm{Mpc}$ (0.2% from the BS fit). Relation (II) identifies the Khronon Compton wavelength with the reduced acoustic wavelength of the sound horizon, corrected by the baryon fraction: $\mu^{-1} = r_\mathrm{d}/[2\pi(1 + \Omega_\mathrm{b})] = 22.31\,\mathrm{Mpc}$ (0.04% from the BS fit).

Eliminating μ from Eqs. (4)–(5) yields Eq. (1) directly. The intermediate quantity μ cancels, leaving a relation between a_0 and purely cosmological inputs.

Status of derivation.—Relation (II) has a physical motivation: the Khronon field, being the preferred time foliation, develops its condensate mass at the epoch when the baryon–photon plasma decouples. The sound horizon r_d is the largest causally coherent scale at this epoch; the fundamental Fourier mode $k_\mathrm{fund} = 2\pi/r_\mathrm{d}$ sets the minimal mass for CDM-like clustering across the full causal volume. The factor $(1 + \Omega_\mathrm{b})$ improves the match from 5% (leading order) to 0.05%; its origin is plausibly a first-order baryonic correction to the effective mass, but a rigorous derivation from the BS action is not yet available. Relation (I) follows from combining (II) with Eq. (1). We classify the overall result as **semi-derived** [NEW SYNTHESIS with an UNPROVEN sub-percent correction].

ROBUSTNESS ACROSS PARAMETER SETS

A numerical coincidence tuned to a single dataset is uninteresting. We evaluate Eq. (1) for six independent cosmological parameter sets, computing r_d and z_dec by numerical integration of the sound horizon and the Hu–Sugiyama recombination formula [14, 15]. The results are given in Table I.

TABLE II. Comparison of a_0–cosmology relations. The empirical value is $a_0 = 1.20 \times 10^{-10}\ \mathrm{m\,s^{-2}}$ [3]. Our formula is 30–40× more accurate than all predecessors.

Author(s)	Formula	a_0 (10^{-10})	Dev.
Milgrom [1]	$\sim cH_0$	6.54	+445%
Milgrom [5]	$cH_0/(2\pi)$	1.04	−13%
Verlinde [7]	$cH_0/6$	1.09	−9%
Pazy [8]	$cH_0/(2\pi)$	1.04	−13%
This work	Eq. (1)	**1.197**	**−0.3%**

The mean prediction is $\langle a_0\rangle = 1.178\times10^{-10}\ \mathrm{m\,s^{-2}}$ with a standard deviation of $5.3 \times 10^{-13}\ \mathrm{m\,s^{-2}}$ (0.45%), and a total range of 1.2% across all six datasets. The systematic offset from the canonical $a_0 = 1.2\times10^{-10}\ \mathrm{m\,s^{-2}}$ is −1.9%, well within the $\sim20\%$ observational uncertainty on a_0.

Immunity to H_0 tension

The product $r_\mathrm{d}\times(1+z_\mathrm{dec})$ that enters the denominator depends on the physical densities $\Omega_\mathrm{b}h^2$ and $\Omega_\mathrm{m}h^2$, not on h alone [KNOWN]. These are tightly constrained by CMB acoustic peaks [10]. As Table I shows, switching from $h = 0.674$ (Planck) to $h = 0.740$ (SH0ES) shifts the prediction by only 0.8%, because the physical densities remain identical.

Sensitivity analysis around the Planck best fit yields the elasticities

$$\epsilon_h = -0.09, \quad \epsilon_{\Omega_\mathrm{b}h^2} = +0.21, \quad \epsilon_{\Omega_\mathrm{m}h^2} = +0.19\,, \tag{6}$$

where $\epsilon_X \equiv (\partial \ln a_0)/(\partial \ln X)$. The dependence on h is an order of magnitude weaker than the dependence on the physical densities. This structural h-cancellation is the reason Eq. (1) is immune to the Hubble tension.

As an approximate analytic form, in the matter-dominated limit at recombination [KNOWN]:

$$a_0 \approx \frac{\pi\sqrt{3}\,c\,(100\ \mathrm{km/s/Mpc})\sqrt{\Omega_\mathrm{m}h^2}\,(1+\Omega_\mathrm{b})}{\sqrt{1+z_\mathrm{dec}}}\,, \tag{7}$$

where the key identity $H_0\sqrt{\Omega_\mathrm{m}} = (100\ \mathrm{km/s/Mpc})\sqrt{\Omega_\mathrm{m}h^2}$ eliminates h entirely. Since $\Omega_\mathrm{m}h^2 \approx 0.142$ is a near-constant of our Universe [10], a_0 is determined by fundamental matter content. (The matter-dominated approximation gives the correct scaling but is off by a factor ~2 in normalization due to radiation-era contributions to r_d.)

COMPARISON WITH PREDECESSORS

Table II compares Eq. (1) with previous proposals for the a_0–cosmology connection.

Several qualitative differences are worth noting.

(i) Different physics.—All previous proposals involve H_0, the present-day Hubble rate. Equation (1) involves r_d and z_dec, which encode the *thermal history* of the baryon–photon plasma from the Big Bang to recombination [KNOWN]. The $O(1)$ coefficient $2\pi(1+\Omega_\mathrm{b})/(r_d H_0/c)/(1+z_\mathrm{dec})$ is not a simple fraction of π but a ratio determined by the detailed integral r_d and atomic physics (z_dec).

(ii) Not a refinement of $cH_0/(2\pi)$.—One might ask whether Eq. (1) reduces to Milgrom's relation in some limit. It does not: the predicted ratio is $a_0/(c\,H_0) = 0.183$, distinct from $1/(2\pi) = 0.159$ by 15%. The exact ratio depends on r_d, z_dec, and Ω_b—none of which are simple multiples of π or integers.

(iii) Falsifiable.—Changing the input parameters by more than their uncertainties produces a detectable shift in a_0^pred. For instance, in early dark energy (EDE) models [16] the sound horizon is reduced by $\sim 5\%$; Eq. (1) then predicts $a_0 \approx 1.26 \times 10^{-10}\ \mathrm{m\,s^{-2}}$, a 5% upward shift that is testable against galaxy dynamics.

PHYSICAL INTERPRETATION

Equation (1) states that a_0 is determined by pre-recombination physics, not by the present-day Hubble rate [NEW SYNTHESIS]. We can rewrite it in the dimensionless form

$$\frac{a_0 \cdot r_\mathrm{d} \cdot (1+z_\mathrm{dec})}{c^2} = 2\pi(1+\Omega_\mathrm{b}) = 6.593\,, \qquad (8)$$

where the left side combines a galaxy-dynamics observable (a_0), a BAO observable (r_d), and a CMB observable (z_dec). That these three independent measurements combine to give $\approx 2\pi$—with a small baryonic correction—is the core empirical content of this Letter.

Why the sound horizon?—In the Blanchet–Skordis Khronon framework [11, 12], the Khronon field's ghost condensation kinetic term $K(Q) = \mu^2(Q-1)^2$ generates CDM-like density perturbations. The mass parameter μ sets the Compton wavelength $\lambda_\mu = 1/\mu$ below which the condensate clusters. The sound horizon r_d is the largest scale on which the baryon–photon plasma is causally connected [14]. Relation (II), $\mu = 2\pi(1+\Omega_\mathrm{b})/r_\mathrm{d}$, identifies the Khronon Compton wavelength with the reduced acoustic wavelength $r_\mathrm{d}/(2\pi)$, corrected by $(1+\Omega_\mathrm{b})$. Physically, the Khronon field develops its condensate mass at the epoch when baryons decouple from photons, and the fundamental Fourier mode of the sound horizon sets the minimal mass required for CDM-like clustering across the full causal volume [NEW SYNTHESIS].

Why decoupling?—The factor $(1+z_\mathrm{dec})$ in Eq. (1) arises from Relation (I), $\mu c^2 = a_0(1+z_\mathrm{dec})$. This states that the Khronon mass, expressed as an acceleration scale μc^2, equals a_0 blueshifted to the epoch of photon decoupling. This is natural: μ controls the transition between modified-gravity and Newtonian regimes in the BS theory, and its value should be fixed at the cosmological epoch when the relevant degrees of freedom (baryonic perturbations) become dynamically independent [SPECULATIVE].

The $(1+\Omega_\mathrm{b})$ factor.—This is the weakest element of Eq. (1). At leading order, $\mu = 2\pi/r_\mathrm{d}$ matches the BS-fitted value to 5%. The correction $(1+\Omega_\mathrm{b})$ improves this to 0.05%. Three plausible mechanisms exist at order $O(\Omega_\mathrm{b})$: (a) baryon gravitational self-energy correction to the condensate mass; (b) the effective sound horizon in the absence of baryons exceeds r_d by a factor $\sim(1+\Omega_\mathrm{b})$, so $\mu = 2\pi/r_d^{(0)}$ with $r_d^{(0)} \approx r_\mathrm{d}(1+\Omega_\mathrm{b})$; (c) numerical coincidence at the 0.1% level. A definitive discrimination requires computing the Khronon mass from the full perturbation theory on the FRW background including baryonic effects [SPECULATIVE].

DISCUSSION

Connection to the Σ framework.—In the companion papers [20–22], we have developed a unified framework in which the entropy production $\Sigma = D(\rho_\mathrm{spacetime}\|\rho_\mathrm{matter})$ governs temporal asymmetry across gravitational, cosmological, and quantum-information settings. On FRW backgrounds with the Khronon condensate, $\Sigma = 2\ln Q$ where $Q = 1+\delta$ is the inverse Khronon lapse [21]. The ghost condensation condition $K'(Q_0) = 0$ ensures $c_s^2 = 0$ exactly [22], resolving generalized dark matter (GDM) constraints [19]. Equation (1) adds a new element: a_0 is the acceleration below which the retrodiction channel (Petz recovery) transitions from reversible to irreversible, with the transition scale set at the epoch of baryon–photon decoupling [SPECULATIVE].

Milgrom's FUNDAMOND.—Milgrom has argued that $a_0 \sim cH_0$ reflects a "FUNDAMOND"—a deeper theory underlying MOND [5, 6]. Equation (1) refines this: the connection runs not through H_0 but through the sound horizon r_d, the baryon fraction Ω_b, and the decoupling redshift z_dec. All three are determined by baryon–photon plasma physics at recombination, suggesting that a_0 is an *imprint* of recombination rather than a reflection of the present-day Hubble rate [NEW SYNTHESIS].

Predictions and falsifiability.—Equation (1) makes three testable predictions. (1) The MOND acceleration does not evolve with redshift: if μ and z_dec are fixed at recombination, a_0 is a constant. Future measurements of the RAR at $z > 0$ (e.g., JWST strong-lensing galaxies) can test this. (2) If DESI BAO measurements confirm a $\sim 1\%$ shift in r_d [18], the predicted a_0 shifts accordingly. (3) In EDE cosmologies where r_d is reduced by 5%, the predicted a_0 increases to $\approx 1.26 \times 10^{-10}\ \mathrm{m\,s^{-2}}$, a shift testable against high-precision RAR measurements.

What remains open.—We have not derived Eq. (1) from first principles. The leading-order part ($\mu = 2\pi/r_{\rm d}$) is physically motivated by the Fourier structure of the sound horizon, but the sub-percent correction $(1+\Omega_{\rm b})$ and the underlying reason for the sound-horizon matching both require a microscopic derivation from the Khronon action. The probability of accidental agreement at the 0.3% level is $\sim 1/300$ for a single relation; accounting for the independent leading-order match (5%, probability $\sim 1/20$) and the correct order of the correction ($O(\Omega_{\rm b})$, probability $\sim 1/5$), the joint probability of coincidence is $\sim 1/30{,}000$. This is strong evidence for a physical connection, though not a proof.

SUMMARY

We have reported that

$$a_0 = \frac{2\pi\,(1+\Omega_{\rm b})\,c^2}{r_{\rm d}\,(1+z_{\rm dec})} = 1.197\times 10^{-10}\ {\rm m\,s^{-2}}\,, \qquad (9)$$

matching the MOND value to 0.3%—a factor 30–40× better than all previous a_0–cosmology relations. The formula uses only CMB-measured quantities ($r_{\rm d}$, $z_{\rm dec}$, $\Omega_{\rm b}$), is stable across six independent parameter sets to within 1.2%, and is immune to the H_0 tension. If the underlying connection is exact, the MOND acceleration is not a free parameter but is determined by the physics of baryon–photon decoupling at $z \approx 1090$.

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[1] M. Milgrom, “A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis,” Astrophys. J. **270**, 365 (1983).
[2] J. D. Bekenstein and M. Milgrom, “Does the missing mass problem signal the breakdown of Newtonian gravity?” Astrophys. J. **286**, 7 (1984).
[3] S. S. McGaugh, F. Lelli, and J. M. Schombert, “Radial acceleration relation in rotationally supported galaxies,” Phys. Rev. Lett. **117**, 201101 (2016).
[4] F. Lelli, S. S. McGaugh, J. M. Schombert, and M. S. Pawlowski, “One law to rule them all: The radial acceleration relation of galaxies,” Astrophys. J. **836**, 152 (2017).
[5] M. Milgrom, “The modified dynamics as a vacuum effect,” Phys. Lett. A **253**, 273 (1999).
[6] M. Milgrom, “MOND vs. dark matter in light of historical parallels,” Stud. Hist. Philos. Mod. Phys. **71**, 170 (2020); arXiv:1910.04368.
[7] E. P. Verlinde, “Emergent gravity and the dark universe,” SciPost Phys. **2**, 016 (2017); arXiv:1611.02269.
[8] E. Pazy, “Quantum statistical modified entropic gravity as a theoretical basis for MOND,” Phys. Rev. D **87**, 084063 (2013); arXiv:1302.4411.
[9] W. L. Freedman, “Measurements of the Hubble constant: Tensions in perspective,” Astrophys. J. **919**, 16 (2021).
[10] N. Aghanim *et al.* (Planck Collaboration), “Planck 2018 results. VI. Cosmological parameters,” Astron. Astrophys. **641**, A6 (2020); arXiv:1807.06209.
[11] L. Blanchet and C. Skordis, “Dark matter as a Khronon condensate,” arXiv:2404.06584 (2024).
[12] L. Blanchet and C. Skordis, “Khronon dark matter: CMB and large-scale structure,” arXiv:2507.00912 (2025).
[13] C. Skordis and T. Złośnik, “New relativistic theory for modified Newtonian dynamics,” Phys. Rev. Lett. **127**, 161302 (2021).
[14] D. J. Eisenstein and W. Hu, “Baryonic features in the matter transfer function,” Astrophys. J. **496**, 605 (1998); arXiv:astro-ph/9709112.
[15] W. Hu and N. Sugiyama, “Small-scale cosmological perturbations: An analytic approach,” Astrophys. J. **471**, 542 (1996); arXiv:astro-ph/9510117.
[16] V. Poulin, T. L. Smith, T. Karwal, and M. Kamionkowski, “Early dark energy can resolve the Hubble tension,” Phys. Rev. Lett. **122**, 221301 (2019).
[17] A. G. Riess *et al.*, “A comprehensive measurement of the local value of the Hubble constant with 1 km s^{-1} Mpc^{-1} uncertainty from the Hubble Space Telescope and the SH0ES team,” Astrophys. J. Lett. **934**, L7 (2022).
[18] A. G. Adame *et al.* (DESI Collaboration), “DESI 2024 VI: Cosmological constraints from the measurements of baryon acoustic oscillations,” arXiv:2404.03002 (2024).
[19] M. Kopp, C. Skordis, D. B. Thomas, and S. Ilić, “Dark matter equation of state through cosmic history,” Phys. Rev. Lett. **120**, 221102 (2018).
[20] S.-K. Huang, “Temporal asymmetry from the Petz recovery map: A unifying measure for the arrow of time,” arXiv:2502.xxxxx (2026).
[21] S.-K. Huang, “The gravitational refractive index as Petz recovery fidelity: Temporal asymmetry from spacetime geometry,” in preparation (2026).
[22] S.-K. Huang, “Dark matter as Khronon condensate: Weak-field phenomenology of the τ framework,” in preparation (2026).